EGYPTIAN FRACTIONS AND THE GREEDY ALGORITHM ON ONE PAGE

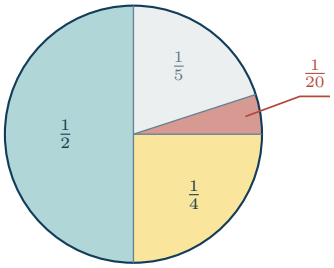
Greedy Tarts

Take the biggest unit-fraction slice that fits, then repeat with the share still owed.

1 / SHARE FOUR TARTS AMONG FIVE PEOPLE

Each plate needs four fifths of a whole tart

A **unit fraction** $1/n$ is one of n equal parts of a whole. An **Egyptian fraction** expresses an amount as a finite sum of **distinct** unit fractions. Here is one person's share:



Colour totals $4/5$; grey is $1/5$.
All fractions refer to a whole tart.

Share still owed	Serve the largest slice	Still owed
$\frac{4}{5}$	$\frac{1}{2}$	$\frac{3}{10}$
$\frac{3}{10}$	$\frac{1}{4}$	$\frac{1}{20}$
$\frac{1}{20}$	$\frac{1}{20}$	0

$$\frac{4}{5} = \frac{1}{2} + \frac{1}{4} + \frac{1}{20}$$

2 / TURN THE SERVING RULE INTO AN ALGORITHM

Largest slice means smallest denominator

For $0 < p < q$ in lowest terms, we want the largest $1/n$ satisfying $1/n \le p/q$. Thus $n \ge q/p$, and the first allowed integer is

$$n = \left\lceil \frac{q}{p} \right\rceil, \qquad R_{\text{new}} = \frac{pn - q}{qn}.$$

The **ceiling** $\lceil x \rceil$ rounds up to an integer. Subtract $1/n$, reduce the remaining fraction, and repeat until zero.

At $3/10$, the ratio $10/3 = 3.33\ldots$ rounds up to 4. Serve $1/4$: it fits, while $1/3$ is too large.

Greedy means taking the largest possible slice now.

3 / A SHRINKING INTEGER PROVES THAT IT STOPS

The numerator falls at every step

The ceiling condition gives

$$n - 1 < \frac{q}{p} \le n \quad \Longrightarrow \quad 0 \le pn - q < p.$$

The new numerator $pn - q$ is smaller than p. Cancelling common factors cannot increase it. A strictly falling sequence of positive integers must end.

So greedy uses **at most the original p terms**. In our example, the reduced numerators fall $4 \to 3 \to 1 \to 0$.

The denominators are distinct too. If the remainder is non-zero,

$$0 < R_{\text{new}} < \frac{1}{n(n-1)} \le \frac{1}{n}.$$

The next slice is smaller, so its denominator is larger.

4 / COMPARE A LOCAL CHOICE WITH THE WHOLE RESULT

Greedy can use more terms

For $9/20$, the greedy recipe gives three pieces:

$$\frac{9}{20} = \frac{1}{3} + \frac{1}{9} + \frac{1}{180}.$$

The remainders are $7/60$, then $1/180$, then 0. Choosing $1/4$ first gives a shorter expansion:

$$\frac{9}{20} = \frac{1}{4} + \frac{1}{5}.$$

Greedy guarantees a finite answer. **Finding the fewest terms** is a separate optimisation problem: the biggest slice now can leave an awkward remainder.

5 / KEEP SPLITTING THE SMALLEST PIECE

One amount, infinitely many recipes

For any integer $n \ge 2$, one unit fraction splits into two:

$$\frac{1}{n} = \frac{1}{n+1} + \frac{1}{n(n+1)}.$$

For example,

$$\frac{1}{3} = \frac{1}{4} + \frac{1}{12} = \frac{1}{4} + \frac{1}{13} + \frac{1}{156}.$$

Split the term with the **largest denominator**. Both new denominators are larger, so no duplicates appear. Repeat to get infinitely many different *finite* expansions of the same positive proper fraction.

YOUR TURN / ONE SHARE, TWO RECIPES

Find the greedy expansion of $7/12$, then a different two-term expansion.

Check: $\frac{1}{2} + \frac{1}{12}$ is greedy; $\frac{1}{3} + \frac{1}{4}$ is another recipe.

History. Unit-fraction arithmetic was used in ancient Egypt; Fibonacci described this greedy algorithm in 1202.

Sources: David Eppstein, *Algorithms for Egyptian Fractions*; Jukka Pihko (2001), introduction to the greedy algorithm, §1.

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